

Question 1

- a) The primary focus of Artificial Intelligence (A.I.) is a rational action. An ideal intelligent agent acts to accomplish its goal. In the context of A.I., explain what is an intelligent agent. (3 marks)

An intelligent agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.

- b) Cognitive modelling approach is one way used to develop an intelligent system. Before such a system can be developed, a precise and testable theories of the human mind must first be discovered. Suggest **THREE (3)** ways to understand how human minds work. Briefly explain each of them. (3 + 3 marks)

Introspection: Try to catch our own thoughts as they go by

Psychological experiments: Observe a person in action.

Brain imaging: Observe the brain in action.

- c) You are required to design an automated medical diagnosis system. Based on this requirement, answer the following questions.
- (i) List **FOUR (4)** important components of the task environment. (4 marks)
- (ii) Provide an example for each based on the solution for **Question 1 c) (i)**. (4 marks)

Performance	healthy patient, reduced cost
Environment	patient, hospital, staff
Actuator	display of questions, tests, diagnoses, treatments
Sensor	touchscreen/voice entry of symptoms and finding.

- d) Briefly explain **THREE (3)** main differences between a *rational agent* and an *omniscient agent*.
(3 marks)

Rational agent

- Doesn't know the actual outcome of its actions.
- Acts under uncertainty most of the time.
- Aims to maximize its performance by achieving the best expected outcome.

Omniscient agent

- Knows the actual outcome of its actions.
- Acts under certainty all of the time.
- Aims to maximize its performance by achieving the best actual outcome.

- e) "*Omniscient agents can be created*".
State your agreement with the statement. Justify your answer. (1 + 4 marks)

No

- Creating omniscient agents is not possible.
- An omniscient agent would be an agent with complete knowledge of all the past, present and future events.
- While advanced AI systems can perform tasks with remarkable efficiency and accuracy, they are still limited by their programming and the data they trained on.
- They are not capable of attaining complete knowledge or omniscience.

Question 2

- a) Problem formulation is an important phase when designing a problem-solving agent. Briefly explain the **SIX (6)** components of a formulated problem. (12 marks)

States	A set of possible states that the environment can be in, referred to as state space.
Initial state	A state that the agent starts in.
Goal state	A state that represents or satisfies the goal.
Action	A set of possible actions available to the agent in a particular state.
Transition model	A description of what each action does and the result.
Cost function	The cost of applying an action in state to reach another state.

- b) **Figure 1** shows interconnected cities in country P where each city is represented by a node, and the distance (in km) between two nodes is represented by an arc connecting them. Assume that you need to analyse all the possible ways and suggest the suitable path to reach the destination, node O from node A. Apply *Breadth-First Search* and *Depth-First Search* to questions below:

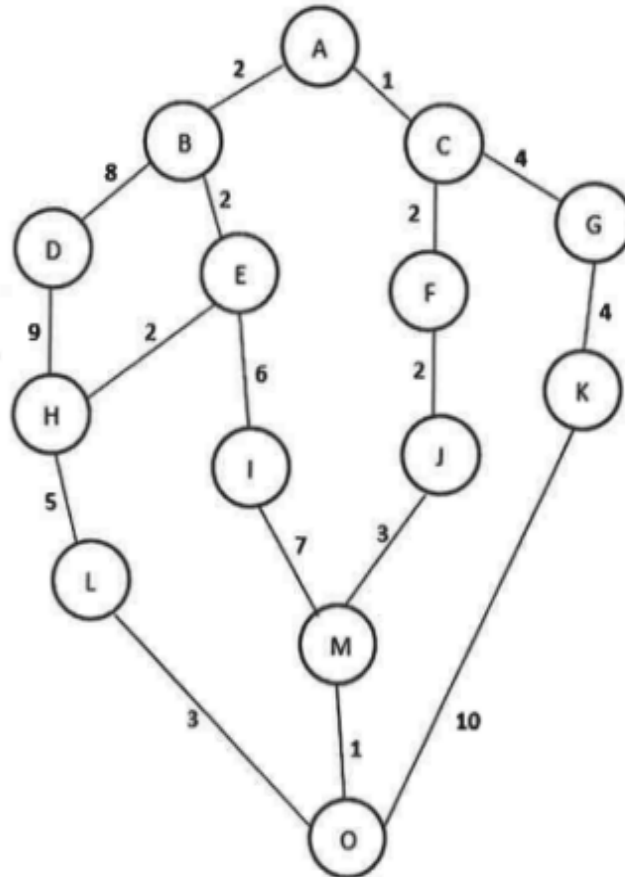


Figure 1 Interconnected cities in country P

Suppose *Breadth-First Search* and *Depth-First Search* are used to solve the problem of searching the shortest path from node A to node O, and both algorithms can deal with repeated states, perform the following tasks for each algorithm.

- (i) Identify the node expansion sequence from the initial state to the goal state in alphabetical order. (2 marks)

- (ii) Identify the solution path. (2 marks)

(iii) Calculate the path cost of the solution. Show the working steps.

(4 marks)

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(iv) Provide a table to interpret the performances of *Breath-First Search* and *Depth-First Search* in terms of completeness and optimality based on the solution for **Question 2 b)**. (5 marks)

	BFS	DFS
Completeness	Complete, reached O	Complete, reached O
Optimality	Optimal, least number of steps	Not optimal

Question 3

a) A game can be formally defined with six elements.

(i) Briefly explain **THREE (3)** problem formulations.

(6 marks)

Initial State	how the game is set up at the start
Action	the set of legal moves in state
Result	the result of each action
Move	the player whose turn it is to move in state
Terminal test	-States where the game has ended
Utility	also known as objective function or payoff function, applies to terminal state to say who won and what is the final score.

- (ii) Calculate the *minimax value* for the nodes (A - F) in the game tree shown in **Figure 2**. (6 marks)

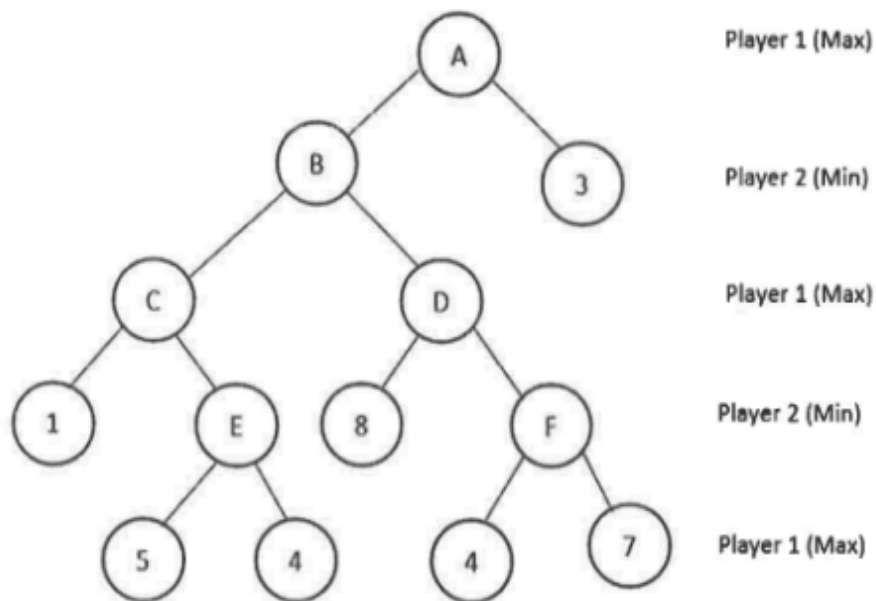
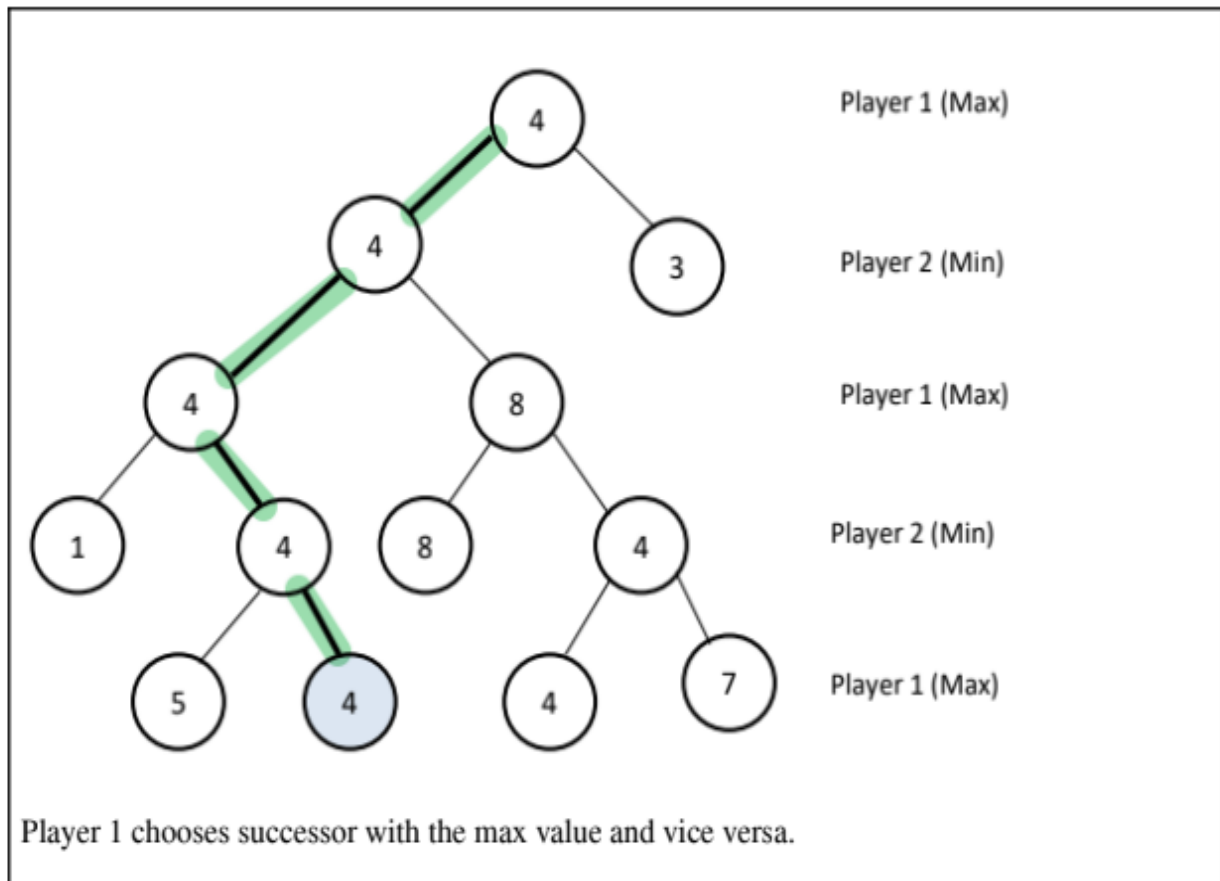


Figure 2 Game tree

A: 4
B: 4
C: 4
D: 8
E: 4
F: 4

- (iii) Identify Player 1's *minimax decision*. To answer this question, you are required to redraw the entire game tree provided in **Figure 2** and highlight the path. Justify the answer. (3 marks)



(iv) Explain why players *MAX* and *MIN* are named so?

(2 marks)

- **MAX** is named so because it tries to maximize the utility/score and achieve the best outcome.
- **MIN** is named so because it tries to minimize the utility/score for MAX, achieving the best outcome for itself.

b) A game can be formally defined with the following elements:

INITIAL STATE, S_0 .

ACTIONS (s)

TO-MOVE (s)

RESULT (s,a)

IS-TERMINAL (s)

UTILITY (s, p)

(i) Provide a brief description to each of the elements above for a *Tic-Tac-Toe* problem.

(7 marks)

c) *Cut-off search* is a useful technique for balancing the computational complexity of playing game algorithm. However, there are limitations in this search algorithm. Identify **ONE (1)** limitation of *cut-off search*.

(1 mark)

Question 4

- a) Briefly define the *Constraint Satisfaction Problem (CSP)*.

(2 marks)

- b) Suppose that, we are given a map of X as in **Figure 3** that shows each of its states and territories. Then, we are given a task of colouring each region with either red, green or blue, in such a way that no neighbouring regions have the same colour.

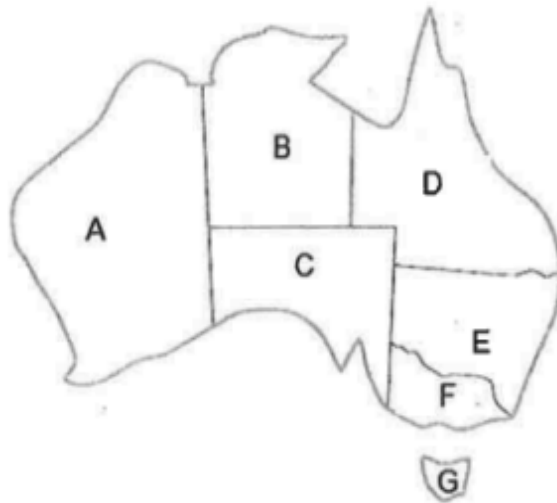


Figure 3 Map of X

- (i) Suppose the colouring task is formulated as an *CSP*, identify its *variables*, *domains* and *constraints*. (6 marks)

- (ii) Based on the answer provided in **Question 4 b) (i)**, identify **ONE (1)** possible solution to this problem. (2 marks)

(iii) "*A solution to a CSP is a consistent and a complete assignment*".

Briefly describe this statement.

(2 marks)

c) Briefly explain **TWO (2)** types of *local consistency*.

(5 marks)

d) Briefly describe *global constraint*. Then, explain why a *global constraint* such as *Alldiff* is still preferred over a set of *binary constraints*.

(4 + 2 marks)

e) Give **ONE (1)** example of game that needs to satisfy an *Alldiff* constraint. Then, briefly describe the game.

(1 + 1 marks)

